

Information sheet

Evidence Without Conclusion: Localising Failure in Autonomous LLM Agent Systems

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1. What is the main claim of the paper? Why is this an important contribution to the autonomous agents and multi-agent systems literature?

Claim. In a multi-agent system whose agents choose their own measurements, hold explicit beliefs and may communicate at will, competence at detecting an environmental change decomposes into two separable capacities: *acquisition* — did the system gather evidence sufficient to support the conclusion — and *inference* — did it convert that evidence into the correct belief. We show these can be measured independently, and that in societies of LLM-based agents acquisition is intact while inference fails almost totally. The agents’ own measurement policy outperforms a fixed reference schedule covering every instrument in the environment, yet one agent of 276 concludes the environment’s law has changed. Adding communication does not repair the failure; it opens a channel that carries unsupported claims while carrying no network structure.

Why it matters here. Three standing assumptions in multi-agent systems research are at stake.

Evaluation. Agent systems are scored on outcomes — conclusion reached, action taken, task completed. An outcome score cannot distinguish an agent that failed to gather what it needed from one that held what it needed and reasoned wrongly, and the remedies are opposite: better tool use or a longer horizon fixes the first and does nothing for the second. We contribute a decomposition that separates them, and offer it as a reusable pattern for agent testbeds rather than only as an analysis of these runs.

Communication as a dependent variable. Multi-agent architectures overwhelmingly schedule, route or require communication, which makes one question structurally unobservable: would these agents communicate at all? Removing the scheduler turns initiation, silence and channel content into measured quantities.

What a channel transmits. Connectivity and epistemic quality come apart — a population can produce communication without producing a network, and still transmit unsupported claims efficiently. That bears directly on trust, norm and reputation modelling in agent societies.

2. What is the evidence you provide to support your claim? Be precise.

Environment. Meridian is a deterministic settlement simulated over thirty days. Measurement noise is keyed by (world seed, instrument, trial index), so evidence is a fixed property of the world and any society can be re-run against an identical universe — every paired analysis depends on this. Physics is fictional, preventing shortcuts from memorised terrestrial physics, and one instrument class is insensitive to the manipulated constant by construction, supplying a built-in negative control. Each site hosts one instrument of each class, so the stream supports causal discrimination, not merely detection. The intervention is covert — a approx. 0.82% shift on day 12, power-analysed before either study — and no prompt or schema mentions anomalies, interventions or simulation; prompt builders structurally accept only a whitelisted view type, and all 235 runs audited clean for leakage.

Design. 235 seeded runs across two studies. Study 2 (85 runs, five arms, societies of two and eight) is pre-registered and frozen at a tagged commit with world seeds quarantined in code until the freeze; seven pre-registered hypotheses (nine rows counting sub-hypotheses) and an eighteen-row decision table were fixed in advance, and no endpoint was computed until every arm was complete. All verdicts are reported, including two hypotheses not supported and one pre-registration conflict recorded rather than amended.

Acquisition is intact. A non-LLM sequential change-point detector, given only per-instrument (day, value) pairs — not the shift magnitude, onset day, world constants, or which instrument class carries the signal — finds the shift in **42 of 42** gravity-shift runs, and in 99–100% of draws subsampled to a two-agent budget, so this is not a benefit of scale. Against a fixed reference schedule over every instrument at six trials daily, the policy gap is *negative* in every eight-agent arm (−0.95 to −1.19): the agents’ own choices supported a **stronger** signal than the reference policy. A scripted mock society applying textbook sequential statistics diagnoses the change in 10 of 10 gravity worlds, establishing solvability without the detector — though not that an autonomous agent should reach 10 of 10.

Inference fails. One agent of 276 concluded a physical law had changed. The rate does not move anywhere along a monotone gradient in available signal from $|z|$ approx. 3.3 to 7.5 across nine conditions — a gradient rather than a single sufficiency claim, so that “the threshold must lie above the level tested” is unavailable. The dominant class in gravity worlds is instrument malfunction or measurement error in all 42 runs. Two routes to the same endpoint separate by model: a *generation* failure (the hypothesis appears anywhere in 2 of 20 final states) and a *commitment* failure (it enters 15 of 20 trajectories, peaks at $p = 0.85$, and is abandoned). Removing the prompt’s mundane-explanation instruction converted conservatism into false alarms, not insight.

Communication is absent when not scheduled. Across **7,680 agent-days** of voluntary opportunity, homogeneous societies produced **zero** voluntary communications — two society sizes, with and without a public bulletin, observed three independent times; zero initiations in 256 agent-runs, exact 95% CI [0, 1.43%]. Not deliberation ending in refusal: of 180 recorded action reasons, exactly one even names the bulletin or a colleague, and that one is a false-positive match.

A catalysed channel carries claims, not structure. One agent from a different model family produces communication in every run, but cascade depth is exactly 1.000 in both catalysed arms — nothing travels a second hop. Of twenty deliveries of unsupported claims originating with the catalyst, **eighteen were incorporated into peers’ beliefs and none challenged**. Attribution is citation-primary: seventeen of twenty-one rest on the recipient citing the delivery event id, and a no-attribution-by-proximity rule discarded 745 belief changes — the rule cut against the finding. Belief dispersion in catalysed arms fell *more slowly* than in the silent counterfactual on identical worlds.

What we do not claim. That agents under-use evidence is established elsewhere at larger scale; our contribution is its location, not the phenomenon. We do not identify the ceiling’s cause, only make the two easiest explanations untenable. Mixed-population results rest on one composition ratio and one model pairing, and we state a serious alternative reading of the contamination result along with the control that would settle it.

3. What papers by other authors make the most closely related contributions, and how is your paper related to them?

STOCKTAKE (Deb & Krishnan, 2026) is the closest methodologically: it evaluates agents against a “fair oracle” driven on the identical observation stream the agent receives. The move is ours; the joint differs. STOCKTAKE isolates *detection* $\rightarrow$ *action*; we isolate *detection* $\rightarrow$ *belief*, which is harder to attribute because no action reveals it. Their oracle uses the environment’s true generative parameters; our detector receives none.

Ríos-García et al. (2026), with Bisht et al. (2026), document evidence non-uptake in 68% of traces across eight domains and 25,000+ runs. We take this as our starting point, not our finding: neither changes the environment’s rules mid-run, computes the strength of the evidence collected, nor separates failure to detect from failure to interpret.

Gandhi et al. (2025), Jansen et al. (2024), Wiemann et al. (2026), Zheng et al. (2026) place agents in worlds discovered by experiment. All are single-agent, and in all the world’s law is non-canonical *from the start* — a task-generation device, not a covert change during a run in which agents have already built a correct theory.

Yan et al. (2026), the field’s communication-centric survey, describes no system in which agents autonomously decide whether to communicate; where the affordance exists (Yang et al., 2024, 2026) no take-up or silence rate is reported. This is why our zero baseline is unmeasured rather than unreported.

Qu et al. (2026) and Cho et al. (2025) quantify claim adoption, but dyadically — no network, no persistent belief state, no provenance. CoSim (Lin et al., 2026) draws the incorporate-versus-challenge distinction we also draw, but injects misinformation exogenously, without provenance tracking or cascade depth. Hu and Qu (2026) we treat as a caution against our own reading rather than support for it. Zhou et al. (2025), whose audit finds 90.7% of LLM-society studies violate at least one of six validity principles, we treat as framing rather than as an incoming objection. Jamshidi et al. (2026) report attenuation rather than amplification along agent chains, which is why we make no amplification claim.

4. Have you published parts of your paper before, for instance in a conference? If so, give details of your previous paper(s) and a precise statement detailing how your paper provides a significant contribution beyond the previous paper(s).

No part has been published in an archival venue — no conference, workshop or journal — and no part is under consideration elsewhere. Two non-archival deposits are disclosed in full.

(a) **Study 1 preprint.** *Observer Zero: Autonomous LLM scientists detect changes to their world but fail to conclude that it changed.* Zenodo 2026, concept DOI 10.5281/zenodo.21872780, CC BY 4.0. Self-deposited, not peer reviewed, not published in any venue. It reports Study 1 only: two-agent societies across model tier, provider and a prompt ablation. The present paper cites it as prior work and draws on four of its findings where they carry the argument. New here: the entire pre-registered Study 2, the acquisition/inference decomposition and detector benchmark, the evidence gradient spanning both studies, the voluntary-communication baseline at both society sizes, the provenance-checked claim-propagation analysis, and the cross-study synthesis. This paper also corrects and records an erratum in that deposit.

(b) Combined preprint. *Observer Zero: Do LLM Agents Form Epistemic Communities?* Zenodo, 12 August 2026, concept DOI 10.5281/zenodo.21906653, CC BY 4.0. A self-deposited earlier version of this manuscript under a title and framing since abandoned. Per Springer Nature policy preprint posting is not prior publication; we disclose it with DOI and licence, and will update the record with the JAAMAS citation on acceptance. This version is reframed around the acquisition/inference decomposition and corrects four numeric errors present in that deposit (7,680 not 6,880 agent-days; 42 of 42 not 40 of 40 detector runs; cascade depth scoped to the catalysed arms; 167 not 190 bulletin reads attributable to one role), each falsifiable from the paper’s own tables.

(c) Software and data. The Meridian platform is deposited in the CoMSES Computational Model Library (release 1.0.0); raw run artifacts are deposited on Zenodo. Deposits, not publications.